

SYSTEM ADMINISTRATION PLAN

Portfolio Builder 2

1. SYSTEM ADMINISTRATION OVERVIEW

The Campus Reserve system is a web-based reservation and facility management platform designed to support efficient booking and scheduling of campus facilities. This system will allow administrators to manage facility availability, monitor reservation requests, and maintain secure user access. The system also supports user registration, reservation submission, approval workflows, and administrative monitoring of resource usage across the campus.

The purpose of this plan is to define the core administrative operations required to ensure the system remains secure, reliable, accessible, and well maintained throughout its lifecycle. As a system administration project, it must address both technical support requirements and governance practices that help maintain operational continuity. The plan is structured to guide how the platform will be managed after deployment, including user administration, policy enforcement, troubleshooting, backups, documentation, and routine maintenance.

The system will be managed by the Capstone project team, with responsibilities shared among the project owner, system administrator, and faculty adviser or technical supervisor. The operational structure ensures that tasks such as account management, system monitoring, backup maintenance, and issue escalation are clearly assigned. This reduces the risk of uncertainty, delays, or unauthorized system changes.

2. ORGANIZATIONAL AND ADMINISTRATIVE PLANNING

The organizational structure for the Campus Reserve system is designed to support clear accountability and effective oversight. The project team acts as the operating entity responsible for system administration and maintenance. The administrator is responsible for configuring and monitoring the environment, managing user accounts, maintaining security controls, and ensuring that the system continues to function according to expected standards.

The project owner or faculty adviser provides strategic oversight and approves major changes that affect the environment, policies, or scope of the system. This governance model supports structured decision-making while keeping operational tasks delegated to the responsible technical personnel. Because the system is used for campus operations, it is important that administrative roles are clearly defined to avoid confusion over who can approve changes, manage accounts, or respond to incidents.

The administrative planning process includes the establishment of procedures for access control, system updates, incident escalation, service continuity, and documentation. The system administrator is expected to maintain a secure environment by enforcing authentication

requirements, managing permissions, monitoring use, and ensuring that backup and recovery procedures are followed. In addition, the administrator must maintain basic operational records, such as user access updates, system changes, and troubleshooting logs.

For this project, administrative responsibilities will be divided into functional areas:

- User account management and access authorization
- System configuration and environment monitoring
- Security maintenance and policy compliance
- Data backup, restoration readiness, and recovery checks
- Issue logging, resolution, and escalation
- Documentation and knowledge management

This approach ensures that Campus Reserve is not only developed as a functional application but also managed as a controlled and secure system. The administrative plan supports continuity, accountability, and operational readiness for both routine use and unexpected technical issues.

3. ORGANIZATIONAL IT POLICIES

3.1 Policy Name: User Access Control Policy

Purpose: To ensure that only authorized individuals are allowed to access the Campus Reserve system and its administrative functions.

Scope: This policy applies to all users, including students, faculty, staff, administrators, and project team members who interact with the system.

Policy Statement: All users must authenticate using approved credentials before accessing the system. Access rights must be assigned according to user roles and operational need. The system administrator must review access requests, manage account permissions, and remove or restrict access when it is no longer necessary.

Responsible Stakeholders: System Administrator, Project Owner, Faculty Adviser.

3.2 Policy Name: Password and Authentication Policy

Purpose: To reduce the risk of unauthorized access by enforcing secure authentication and strong password practices.

Scope: This policy applies to all user accounts and administrative accounts associated with Campus Reserve.

Policy Statement: Users must maintain strong passwords, protect their credentials, and comply with account security requirements. Multi-factor authentication, password verification, and secure account recovery procedures should be implemented wherever applicable. Weak, shared, or compromised credentials must be reset immediately and reported to the system administrator.

Responsible Stakeholders: System Administrator, Users, Project Team.

3.3 Policy Name: Data Privacy and Confidentiality Policy

Purpose: To protect personal information, reservation records, and operational data stored in the system.

Scope: This policy applies to all information captured, processed, and stored within Campus Reserve.

Policy Statement: System data must be collected and used only for authorized purposes. Sensitive information should be restricted to individuals with legitimate access requirements and must not be disclosed to unauthorized users. The system administrator is responsible for enforcing confidentiality and ensuring that all data handling follows security best practices.

Responsible Stakeholders: System Administrator, Project Owner, Faculty Adviser, Users.

3.4 Policy Name: System Change Management Policy

Purpose: To ensure that changes to the system are planned, evaluated, and controlled before implementation.

Scope: This policy applies to software updates, configuration changes, feature additions, and maintenance activities affecting the application or its environment.

Policy Statement: All major changes must be documented, tested, and approved before deployment. Unapproved or undocumented alterations may create operational risk, disrupt facility booking services, or compromise system reliability. Proper change control helps maintain stability and ensures accountability for updates.

Responsible Stakeholders: System Administrator, Project Owner, Faculty Adviser, Development Team.

3.5 Policy Name: Incident Reporting and Response Policy

Purpose: To ensure that system issues, security concerns, and service disruptions are reported and managed in a consistent and timely manner.

Scope: This policy applies to login issues, account problems, system errors, reservation failures, data concerns, and any other operational incidents affecting the platform.

Policy Statement: All incidents must be documented and reported to the responsible administrator or project lead. Serious issues must be escalated quickly, and corrective actions must be recorded for future review and improvement. A structured response process supports faster resolution and helps reduce repeat issues.

Responsible Stakeholders: System Administrator, Project Owner, Faculty Adviser, Users.

4. ADMINISTRATIVE POLICIES

4.1 Policy Name: Account Creation and Maintenance Policy

Purpose: To ensure that user accounts are created, updated, and removed in a secure and consistent manner.

Responsible Personnel: System Administrator, Project Owner.

Administrative Guidelines: The administrator must verify the identity of each new user before creating an account. Each account should be assigned the correct role and appropriate access permissions. When a user changes role, status, or eligibility, the administrator must update or disable the account accordingly. Account changes must be documented to maintain accountability and support auditing.

4.2 Policy Name: Backup and Recovery Policy

Purpose: To protect system data and maintain continuity in the event of data loss, failure, or accidental deletion.

Responsible Personnel: System Administrator, Project Owner.

Administrative Guidelines: Backup procedures must be performed according to an established schedule and should cover critical reservation records and application data. Backup files must be stored securely and verified periodically to ensure they are complete and recoverable. The administrator should record backup dates, storage locations, and restoration test outcomes to support operational continuity.

4.3 Policy Name: Security Monitoring and Review Policy

Purpose: To maintain system integrity by monitoring activity and identifying risks before they become serious issues.

Responsible Personnel: System Administrator, Faculty Adviser.

Administrative Guidelines: The administrator should review login activity, system logs, failed authentication attempts, and access alerts on a regular basis. Any suspicious or unauthorized behavior must be investigated promptly, documented, and escalated when necessary. Periodic security reviews should also be performed to identify weaknesses and recommend corrective actions.

5. STANDARD OPERATING PROCEDURE (SOP)

SOP Title: Password Reset and Account Security Procedure

Purpose:

The purpose of this SOP is to provide a consistent and secure method for handling password reset requests and account security concerns in Campus Reserve. This procedure helps maintain user trust, reduce unauthorized access, and support efficient issue resolution.

Scope:

This SOP applies to all users who require password recovery, password changes, or account security assistance, including both general users and administrative personnel.

Responsible Personnel:

- System Administrator
- Project Owner or Faculty Adviser (when escalation is required)

Required Resources:

- User account management interface
- Password reset or verification workflow
- Authentication and security validation tools
- Access log or audit documentation
- Incident reporting form or log sheet

Step-by-Step Procedure:

1. Receive the user's request for password reset or account security assistance through the approved support or application process.
2. Verify the identity of the requester using the available authentication checks or security verification steps.
3. Confirm whether the issue is related to a forgotten password, password change request, or suspicious account activity.
4. If the request is valid, issue the password reset or account recovery process according to the system's approved procedure.
5. Require the user to create a strong password that meets the system's security requirements.
6. If the account has signs of compromise or unauthorized activity, immediately review the access logs and restrict suspicious behavior where necessary.
7. Notify the user of the status of the reset or security review and provide any additional security instructions required.
8. Record the request, verification steps, actions taken, and resolution outcome in the system log or administrative documentation.
9. Escalate unusual or high-risk incidents to the project owner or faculty adviser for further review.

Expected Output:

A secure and verified account recovery or password reset process that restores user access safely while documenting the incident and protecting the system from unauthorized activity.

6. DOCUMENTATION STANDARDS

The Campus Reserve system requires clear and organized documentation to support administration, security, maintenance, and governance. Documentation should be maintained consistently so that system administrators, project stakeholders, and support personnel can reference the correct procedures, policies, and records when performing operational tasks.

Documents to be maintained include:

- System Administration Plan
- User account register and access logs
- Password reset and account recovery records
- Backup and recovery logs
- Incident reports and resolution records
- Configuration and version history
- Security review records and access approval documentation
- Maintenance and troubleshooting logs

Purpose of each document:

- The System Administration Plan establishes the overall governance framework for the system.
- User account records track user access, role assignment, and authorized permissions.
- Password recovery records document security-related requests and actions taken.
- Backup records support data protection and disaster recovery readiness.
- Incident logs record system problems, corrective actions, and lessons learned.
- Configuration records help maintain traceability and support future troubleshooting.

Responsible personnel:

- The System Administrator is responsible for maintaining technical and security documentation.
- The Project Owner or Faculty Adviser reviews major updates and approves policy-related changes.
- Team members may contribute documentation when their work affects access, configuration, or system operations.

